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CASE

Chitta Express: Racing against barriers to vaccinate the unreachable

Sujal Jayantibhai Mayatra¹ | Bhawna Priya²

¹ Indian Administrative Service, ² Indian Institute of Management, Raipur, India

INTRODUCTION

In early 2021, as India launched its nationwide COVID-19 vaccination drive, Amit Desai, District Collector of Panchmahal in Gujarat, was confronted with a unique set of challenges. While other regions were making steady progress¹, Panchmahal posed formidable obstacles due to its remote geography and diverse tribal communities. Spread over a vast and uneven terrain, the district residents lived in isolated hamlets with limited access to healthcare. The district's rugged landscape made it difficult to implement a conventional vaccination campaign, especially in areas where roads were either non-existent or impassable.

Adding to the logistical challenges was the presence of deep-rooted vaccine hesitancy among the district's tribal populations. Many residents were mistrustful of government health initiatives, relying instead on traditional healing practices. This mistrust, coupled with widespread misinformation about the vaccine, further complicated efforts to ensure broad participation in the vaccination drive.

As the virus continued to spread, the pressure mounted for Amit and his team to find a solution that could overcome both the logistical and cultural barriers. The standard approach of setting up fixed vaccination centers had proven ineffective in other rural areas, and Amit knew that a more innovative, adaptable strategy would be necessary to protect Panchmahal's vulnerable populations.

¹ Total 5,43,98,707 - 77.95% of citizens received at-least one dose of vaccine in Gujarat, about 5,40,32,774 - 77.42% fully vaccinated. Total 13,20,979 - 55.31% of citizens received at-least one dose of vaccine in Gujarat - Panchmahal, about 13,39,072 - 56.07% fully vaccinated (as on 17th February 2025). Source: <https://vaccinate-india.in/district/185> Accessed on: 17th February 2025

PANCHMAHAL DISTRICT AND THE CHALLENGE OF VACCINATION

Panchmahal, located in the eastern part of Gujarat, spans 5,231 square kilometers ² and is home to approximately 2.4 million people ³ as of 2011 census. The district is administratively divided into seven ⁴ blocks (**Exhibit 1**), with Jambughoda and Ghoghamba being predominantly tribal regions (**Exhibit 2**). These areas, situated approximately 70 kilometers and 35 kilometers ⁵ from the district headquarter Godhra, respectively, are characterized by difficult terrain (**Exhibit 3**) and widely dispersed rural settlements, posing substantial challenges for public health initiatives.

One of the primary hurdles in reaching the district's population stems from its unique geographical and demographic features. The people here primarily rely on subsistence farming and reside in small, isolated hamlets, many of which are tucked away in forested areas and hilly regions. The road infrastructure is limited, with numerous villages accessible only by narrow, unpaved paths. During the monsoon season, these paths become even more difficult to traverse, further isolating these communities ⁶.

In addition to the logistical challenges imposed by the district's geography, the lower literacy rates ⁷ in Panchmahal (70.99%) compared to the Gujarat state average (78.03%) complicate efforts to improve public health awareness. This gap in education exacerbates the spread of misinformation about vaccines, which has led to significant vaccine hesitancy in these tribal areas ⁸. Deep-rooted cultural beliefs and mistrust of government health initiatives have made it difficult to foster confidence in modern medical interventions ⁹.

Jambughoda presents a unique set of challenges due to its dense forests and rich biodiversity ¹⁰. While the forests provide natural beauty and attract eco-tourism, they also create barriers to healthcare delivery. Many villages in this block are located deep within forested areas, and healthcare workers struggle to reach them using standard medical vehicles. Furthermore, the tribal communities in Jambughoda are deeply connected to their traditional healing practices,

² Census of India, 2011 DH_2011_2417_PART_A_DCHB_PANCH_MAHALS.pdf
<https://censusindia.gov.in/census.website/data/handbooks> Accessed on: 17th February 2025

³ Census of India, 2011 https://censusindia.gov.in/census.website/data/data-visualizations/PopulationSearch_PCA_Indicators
 Accessed on: 17th February 2025

⁴ Central Ground Water Board (CGWB): Panchmahal report <https://www.cgwb.gov.in/cgwbpm/publication-detail/169>
 Accessed on: 17th February 2025

⁵ Google. *Google Maps: Distance from Godhra to Jambughoda and Distance from Godhra to Ghoghamba.*
<https://www.google.com/maps>. Accessed on: 17th February 2025

⁶ Gujarat infrastructure development board <https://gidb.org/> Accessed on: 17th February 2025

⁷ <https://www.indiacensus.net/states/gujarat/literacy> Accessed on: 17th February 2025

⁸ UNICEF. *Immunization among Tribal Population in India: A Need Assessment Report.* Ministry of Health and Family Welfare, Government of India.
<https://www.unicef.org/india/media/6716/file/Immunization%20among%20Tribal%20Population%20in%20India.pdf>
 Accessed on: 17th February 2025

⁹ <https://www.ijcmph.com/index.php/ijcmph/article/view/11993> Accessed on: 17th February 2025

¹⁰ Gujarat Forest Department: Jambughoda Wildlife Sanctuary Overview <https://gujaratforest.org/wildlife-jambuhoda1.htm>
 Accessed on: 17th February 2025

which often conflict with modern healthcare interventions, leading to resistance to vaccination programs ¹¹.

Ghoghamba faces similar challenges with its hilly terrain and scattered villages. Many residents live far from main roads, making it difficult to set up centralized healthcare facilities. As in Jambughoda, traditional healing practices are deeply entrenched in Ghoghamba, and mistrust of government initiatives remains a significant barrier. Additionally, the lack of nearby healthcare facilities only exacerbates the challenge of reaching these communities.

CONVENTIONAL STRATEGIES TESTED: THE STRUGGLE TO VACCINATE PANCHMAHAL'S TRIBAL REGIONS

Establishing vaccination centers at schools, health facilities like PHCs and CHCs seemed like a logical step. However, the rural and tribal populations were located far from these facilities, and the poor road infrastructure further exacerbated the problem; particularly in blocks like Jambughoda and Ghoghamba, meant that many villages were accessible only by narrow, unpaved paths. During the monsoon season, these paths often became impassable due to flooding and landslides, cutting off entire communities for weeks at a time. Even outside of the monsoon season, the district's lack of reliable public transportation made it nearly impossible for residents in remote hamlets to travel long distances to vaccination centers.

In many cases, residents would have had to walk several kilometers to the nearest vaccination facility. For vulnerable groups, such as the elderly and those with limited mobility, this was not a feasible option. As a result, the fixed-center approach failed to ensure equitable vaccine distribution across the district, leaving large sections of the population unvaccinated. The logistical challenges of getting healthcare workers and vaccines to these remote areas became a major stumbling block in the district's vaccination efforts.

In addition to the logistical challenges, vaccine hesitancy was a significant barrier, often driven by misconceptions and fear among villagers. Fieldwork revealed several common sentiments expressed by community members, including:

"Why should we waste money traveling to the vaccination center when we are already healthy?"

"There is no need to take the vaccine. It will only make us sick."

"We have survived without vaccines so far; why take the risk now?"

"I heard people get fever and weakness after vaccination. I can't afford to take days off from work."

To address this, Amit and his team considered launching a comprehensive *Information, Education, and Communication (IEC) campaign* aimed at dispelling these fears and raising awareness about the importance of vaccination. The IEC campaign would involve targeted messaging through local media, community meetings, and the distribution of informational

¹¹ Ministry of tribal affairs: Traditional practices in tribal regions <https://tribal.nic.in> Accessed on: 17th February 2025

pamphlets. It would also include engaging local influencers and tribal leaders to act as intermediaries who could help build trust within their communities.

However, while IEC campaigns had the potential to gradually change attitudes and increase vaccine acceptance, they were a long-term solution. Shifting deeply ingrained cultural beliefs and overcoming vaccine hesitancy would take time—time that Panchmahal did not have as the virus continued to spread across the district. The urgency of the situation required an immediate response, and relying solely on an awareness campaign was not feasible. Amit understood that in a region as isolated as Panchmahal, where health infrastructure was limited and traditional beliefs were strong, a different approach would be needed to reach the tribal population effectively and ensure widespread vaccine coverage.

Recognizing that these conventional approaches alone would not suffice, Amit understood the need for a more adaptive and innovative solution—one that could overcome both the geographical isolation and cultural barriers of Panchmahal's tribal communities. This realization led to the birth of the Chitta Express, an agile initiative aimed at taking the vaccine directly to the people, rather than expecting them to come to fixed centers.

CHITTA EXPRESS: REACHING THE UNREACHABLE

The Chitta Express ¹² was designed to navigate the district's challenging terrain and cultural resistance by bringing mobile vaccination units into Panchmahal's most remote and hesitant communities. Unlike conventional strategies, which relied on static centers and long-term awareness campaigns, the Chitta Express represented a dynamic, on-the-ground solution. By utilizing specially equipped motorbikes—dubbed Chitta Bikes—and larger mobile vaccination vans called Sanjeevani Vans, the initiative aimed to reach even the most inaccessible villages nestled within forests and hills. The name “Chitta” was chosen to reflect the speed and agility required to cover difficult terrain quickly and efficiently. The name "Sanjeevani" was inspired by the legendary life-saving herb from Indian mythology

These mobile units were more than just a transportation mechanism for vaccines. The teams operating the Chitta Express were trained not only to administer vaccines but also to engage with local communities, addressing concerns and dispelling myths in culturally sensitive ways. The strategy was if people could not come to the vaccine; the vaccine would go to the people—right to their doorsteps.

At the heart of this operation were the Chitta Bikes (**Exhibit 5**), specially equipped motorcycles that could navigate the narrow, rugged paths of Panchmahal's hilly tribal regions. Each bike was manned by a healthcare worker and a driver, carrying a limited number of vaccines and essential medical supplies. The agility of the Chitta Bikes allowed them to reach villages tucked away in the hills, places no ordinary vehicle could access. The Sanjeevani Vans (**Exhibit 6**), larger mobile units, worked alongside the Chitta Bikes. These vans were used to cover flatter, more accessible regions.

¹² Refer to **Exhibit 4** for Chitta Express branding

CHITTA EXPRESS ROLLOUT: A COLLABORATIVE INITIATIVE

The project was a government-led initiative implemented in collaboration with Care India¹³ (www.careindia.org), a Non-governmental Organization (NGO). The NGO provided financial support. Government backing was secured with the support of Nimishaben Suthar, Minister of State for Health & Family Welfare, Medical Education, and Tribal Development. A total of 20 vehicles, comprising 10 Chitta Bike Express and 10 Sanjeevani Express, were deployed across the district, with the official launch by Nimishaben Suthar on November 18, 2021. All vehicles were equipped with an audio system that broadcasted messages emphasizing the importance of COVID-19 vaccination, encouraging beneficiaries to get vaccinated, and conducting IEC activities within the community and near vaccination sites. Additionally, each vehicle featured distinct branding, effectively conveying its purpose while reinforcing the vaccination message.

Each Chitta Bike was operated by a team of two individuals: a bike rider and a vaccinator. The rider was responsible for serving as a verifier, ensuring that vaccination data was accurately uploaded to the CoWIN portal¹⁴, and assisting in mobilizing eligible beneficiaries. The vaccinator, in addition to administering vaccines, also contributed to mobilization efforts as needed. The Sanjeevani Express was staffed by a driver, a mobilizer, and a vaccinator. Vaccines were transported in a vaccine carrier¹⁵, a compact insulated container designed to maintain the required storage temperature. To facilitate smooth communication and minimize language barriers, the administration coordinated efforts to assign on-ground team members and field staff from the local communities of Panchmahal district.

The operational structure was organized with Care India at the top, followed by state-level liaison officer, and then two district-level coordinators overseeing implementation. The district coordinators were responsible for designing the micro-plan, which outlined the express route based on areas with low vaccination coverage and due beneficiaries. This planning was conducted in collaboration with the Taluka Health Officer (THO) and the Medical Officer in Charge (MOIC) of PHCs. The route plan was communicated to field staff in advance to ensure efficient mobilization and coordination. Vehicles were branded with COVID-19 IEC materials and equipped with a Public Address (PA) system to enhance awareness and outreach efforts. To maintain oversight and ensure smooth execution, a daily evening briefing was conducted under the chairmanship of the District Development Officer (DDO), where progress on COVID-19 vaccination efforts was reviewed, and necessary adjustments were made.

As the project was initiated and supported by the Ministry of Health (State), with government directives instructing authorities to actively involve local influencers and grassroots leaders in the vaccination drive. To enhance credibility and encourage participation, the initiative engaged ASHA workers, Anganwadi workers, Sarpanches (village heads), Gram Panchayat members, local religious leaders, school principals, and other respected community figures. These

¹³ Refer to **Exhibit 7** for Care India social media post related to the project

¹⁴ A digital platform introduced by the Government of India to monitor COVID-19 vaccination drives connecting various stakeholders such as vaccine manufacturers, administrators, and verifiers, public and private vaccination facilities, and vaccine recipients.

¹⁵ Refer to **Exhibit 8** for vaccine carrier

individuals held significant influence within their communities due to their long-standing engagement in social welfare, governance, and public health initiatives. Their endorsement played a pivotal role in addressing vaccine hesitancy, as they were trusted voices among the villagers.

STRATEGIC IMPLEMENTATION OF THE CHITTA EXPRESS

Amit's team conducted a comprehensive survey of the district, identifying the areas with the lowest vaccination rates and highest vaccine hesitancy—predictably, these were the remote tribal blocks of Jambughoda and Ghoghamba. These areas became the focal point for the first phase of the initiative.

Every day, the Chitta Bikes and Sanjeevani Vans followed carefully mapped-out routes, ensuring that no village, no matter how isolated, was not overlooked. Equipped with portable audio systems, the mobile units spread messages about the importance of vaccination in local languages, further building trust with the communities they served. Often, local leaders would join the mobile units, lending their support and helping to sway hesitant residents.

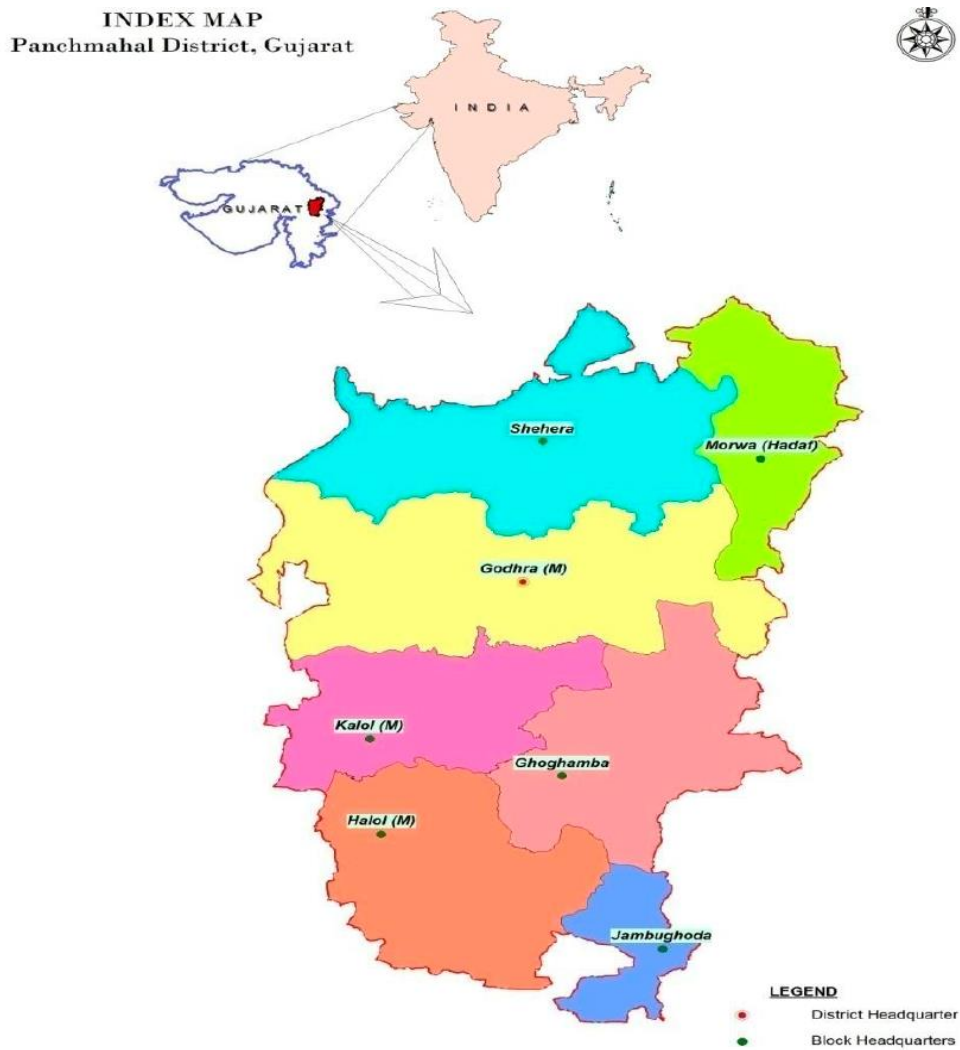
The operation was designed to be agile. Amit and his team held daily briefings to review progress and address any challenges that arose. This real-time feedback allowed for quick adjustments to routes and strategies, ensuring that the initiative stayed on track and vaccination rates continued to rise.

The initiative led to an increase in vaccination coverage in Panchmahal (**Exhibit 9**), particularly in remote tribal areas that had previously been difficult to reach. Beyond improving access to vaccines, it also influenced interactions between healthcare workers and the local population. Villagers who were initially hesitant became more receptive to vaccination efforts, as the Chitta Express teams engaged with them through a personalized and community-focused approach.

SUGGESTED READINGS

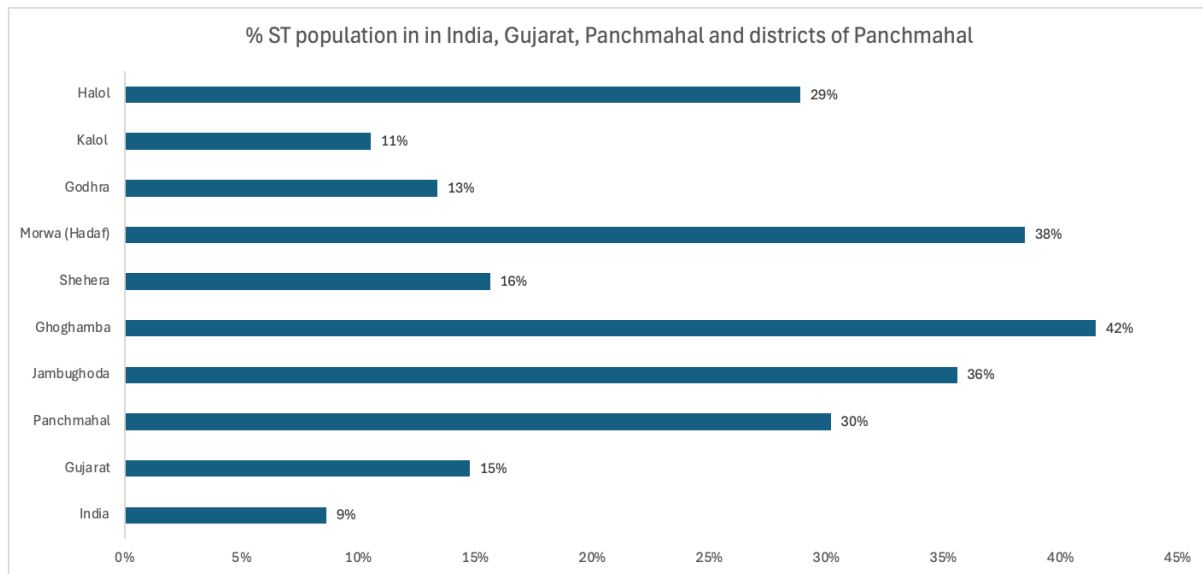
- Piramal Foundation. *Dispelling Vaccine Hesitancy in Tribal Districts in India*. <https://piramalfoundation.org/Docs/Dispelling%20Vaccine%20Hesitancy%20in%20Tribal%20Districts%20of%20India.pdf>. Accessed on: 17th February 2025.
- Ministry of External Affairs, Government of India. *Sentinels of the soil*. <https://www.mea.gov.in/Images/french/story9.pdf>. Accessed on: 17th February 2025.
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Exhibit 1. Map of Panchmahal district comprising of the seven districts



Source: <https://www.cgwb.gov.in/cgwbpm/publication-detail/169>

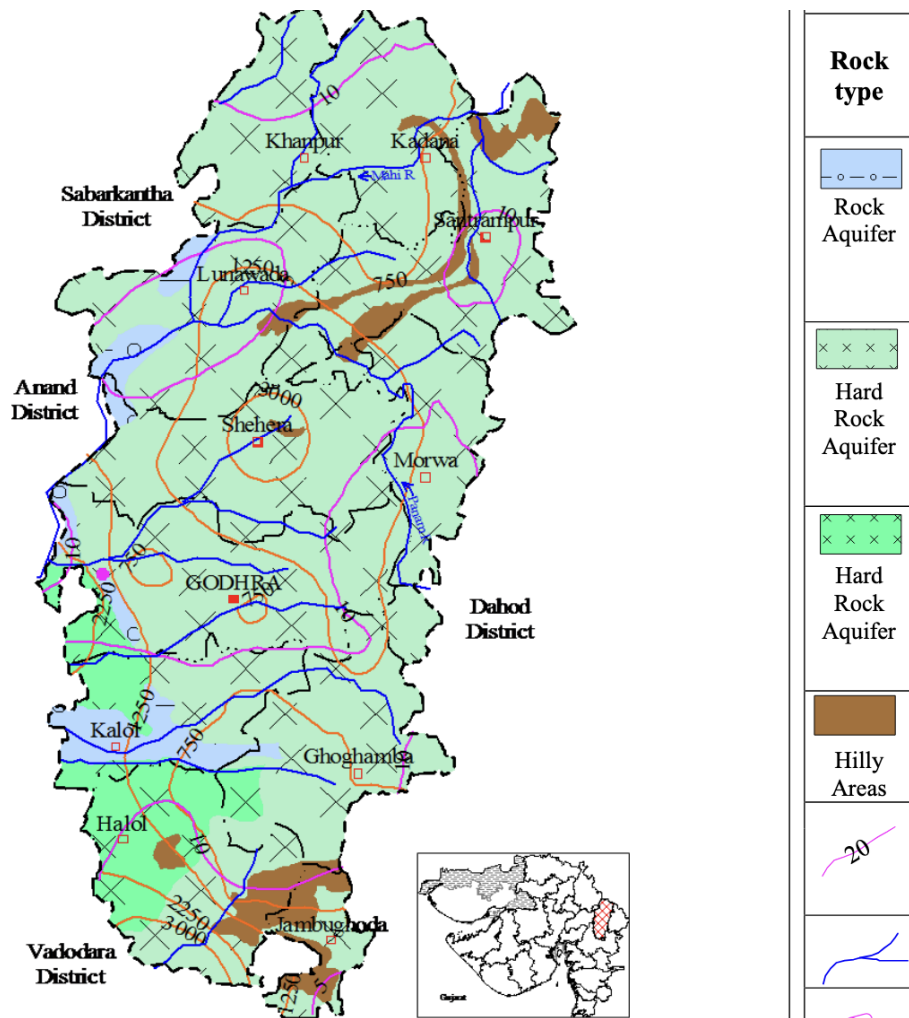
Exhibit 2. Share of Scheduled Tribe (ST)* population in India, Gujarat, Panchmahal, Jambudhoda and Ghoghamba for comparison purpose



* Scheduled Tribes are tribal communities recognized under the Constitution of India for the purposes of socio-legal protection, representation, and targeted welfare schemes.

Source: https://censusindia.gov.in/census.website/data/data-visualizations/PopulationSearch_PCA_Indicators

Exhibit 3. Hydrogeological map of Panchmahal district showing hilly terrain in the district



Source: Hydrogeological Map of Panchmahal District in the Central Ground Water Board (CGWB) report https://cgwb.gov.in/old_website/District_Profile/Gujarat/Panchmahal.pdf. Accessed on: 17th February 2025

Exhibit 4. Chitta express branding (numbers have been put on the image to highlight the details)



1. Hon. Nimishaben Suthar 2. COVID vaccination 3. Cheetah Express Bike 4. Both doses of COVID-19 vaccination are essential for complete protection 5. COVID-19 vaccine is safe 6. District Administration, Panchmahal 7. CARE (organization logo)

Source: Designed by Care India (www.careindia.org) under the supervision of Amit Desai and his teammates

Exhibit 5. Chitta bikes administering COVID vaccines in Panchmahal district



Source: Captured by team members who worked on-site

Exhibit 6. Sanjeevani van flag off (left image) and administering of COVID vaccine (right image) in Panchmahal district



Source: Captured by team members who worked on-site

Exhibit 7. Social media post by Care India on Chitta Express



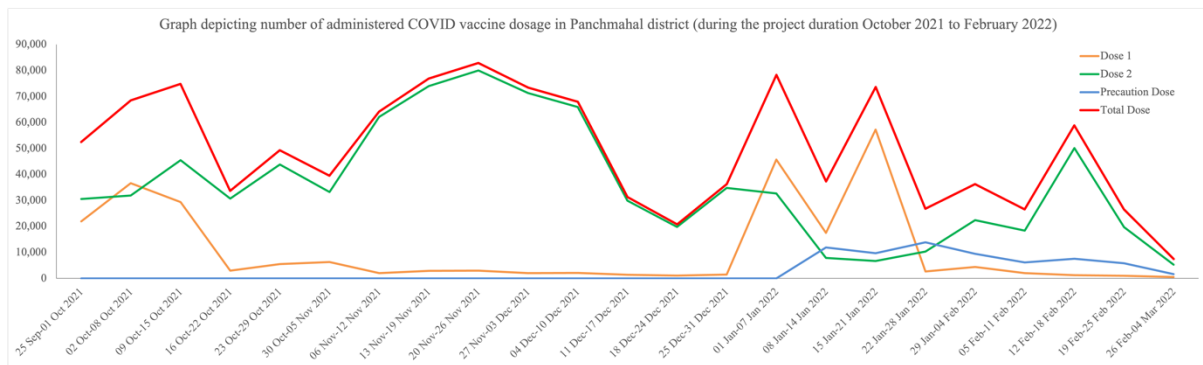
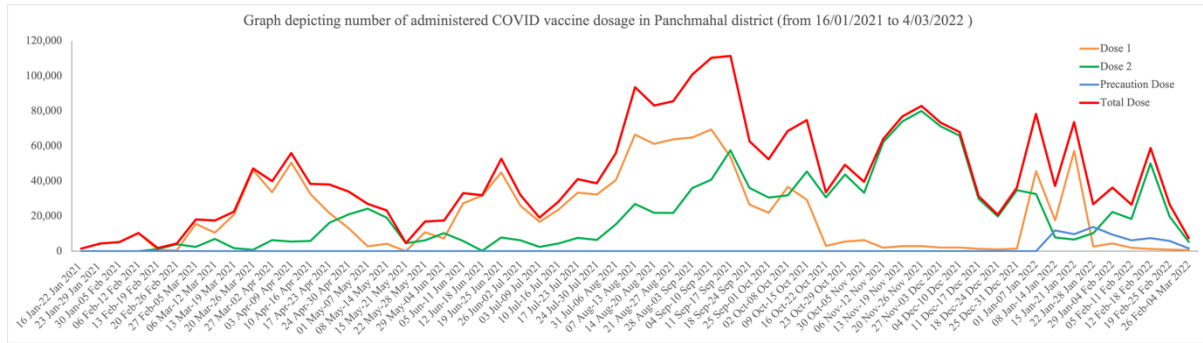
Source: <https://www.facebook.com/100067633087906/posts/10166269896070624/> Accessed on: 17th February 2025

Exhibit 8. COVID vaccine carrier



Source: https://www.business-standard.com/article/current-affairs/coronavirus-update-live-india-corona-cases-covishield-vaccines-covid-19-news-state-wise-121011200162_1.html Accessed on: 17th February 2025

Exhibit 9. Graph depicting total number of administered COVID vaccine dosage in Panchmahal district



Source: CoWIN portal <https://dashboard.cowin.gov.in/>